EXCELLENCE

CLASS 11 - 2026-27

Physics | Motion 01

Motion in a Straight Line

Frame of Reference · Rest & Motion · Distance & Displacement

By - Excellence Programme

20 October 2026

What we learn today

01 Frame of reference and the observer

02 Position → rest and motion (with x–t graphs)

03 Distance vs displacement — definitions and figures

04 Rules: distance ≥ |displacement|

05 Number-line, plane-path and 3-D examples

06 Zero cases, square park, board MCQs

How to study this lecture

Every idea comes with a picture. First look at the figure, then read the boxed rule, then try the green solution steps without peeking.

We start Motion in a Straight Line after Vectors. Today is only foundations: frame, rest–motion, distance and displacement. Speed and acceleration come later.

Class notes come after teaching. These slides are for the board.

Why “distance of Delhi?” fails

Someone asks: “What is the distance of Delhi railway station?” Without saying from where we measure, the question has no meaning.

Every length needs a starting place — a frame of reference

- Wrong question → reference not given
- Better: distance of Delhi from Patna along a named path



Frame of reference — the stand we stand on

A frame of reference is the place (and the axes) from which we take the observation. The observer is the person who takes that observation.

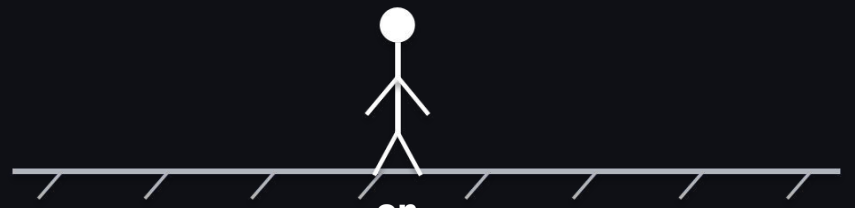
Lift = frame



observer

Observer rides with the lift

Ground = another frame



on ground

Rough work / extra work

Bus at 25 m/s — who is at rest?

The bus moves to the right at 25 m/s relative to the ground. Passengers A, B, C sit inside. Person D stands on the ground.



A is at rest with respect to A. Observer always assumes he is at rest w.r.t. himself.

Rest and motion are relative

Relative to passenger A, B may be at rest (both sit still in the bus). Relative to ground observer D, both A and B move at 25 m/s with the bus.

- Same object can be “at rest” in one frame and “in motion” in another
- Never say rest or motion without naming the frame

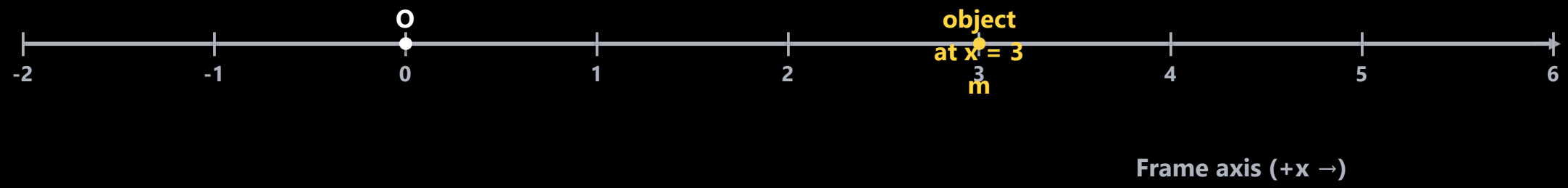
Nothing is absolute for rest and motion in this universe

“When will Delhi come?” — from the bus frame, Delhi approaches you. Physics allows that language.

Position

Position means the location of an object with respect to a chosen frame of reference. On a number line we write it as a coordinate (with a unit).

Position = location of object w.r.t. the frame



Rest and motion from position

Watch the position as time passes in the same frame.

Rest

Position does not change with time.
x stays the same.

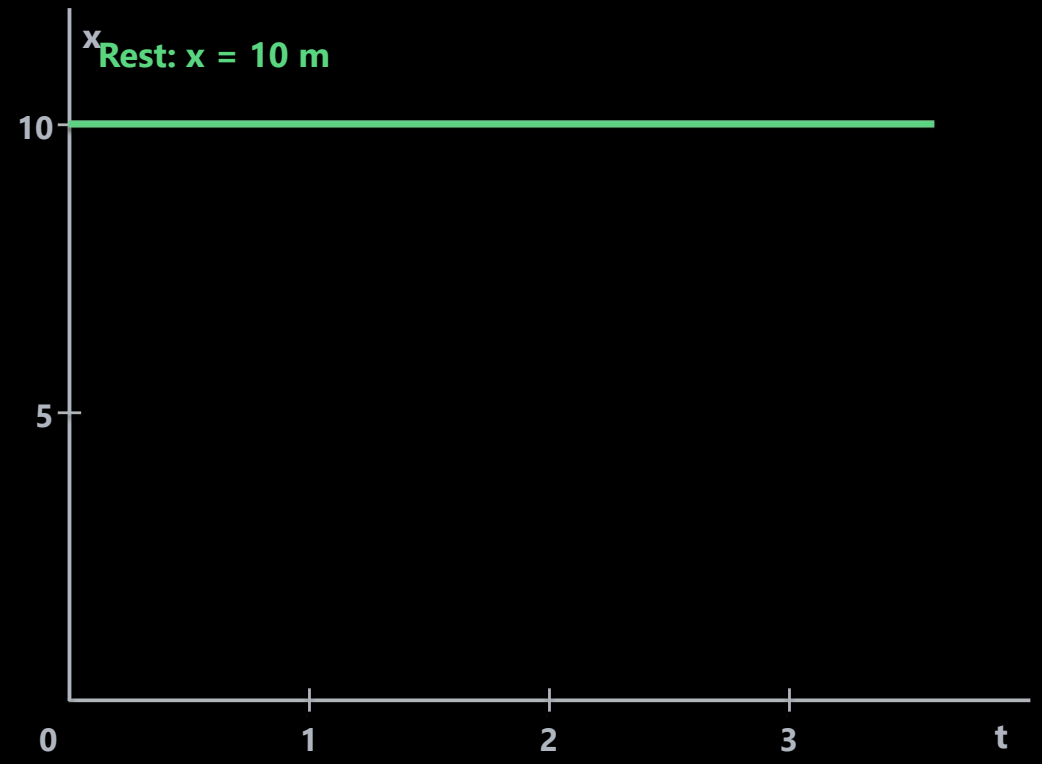
Motion

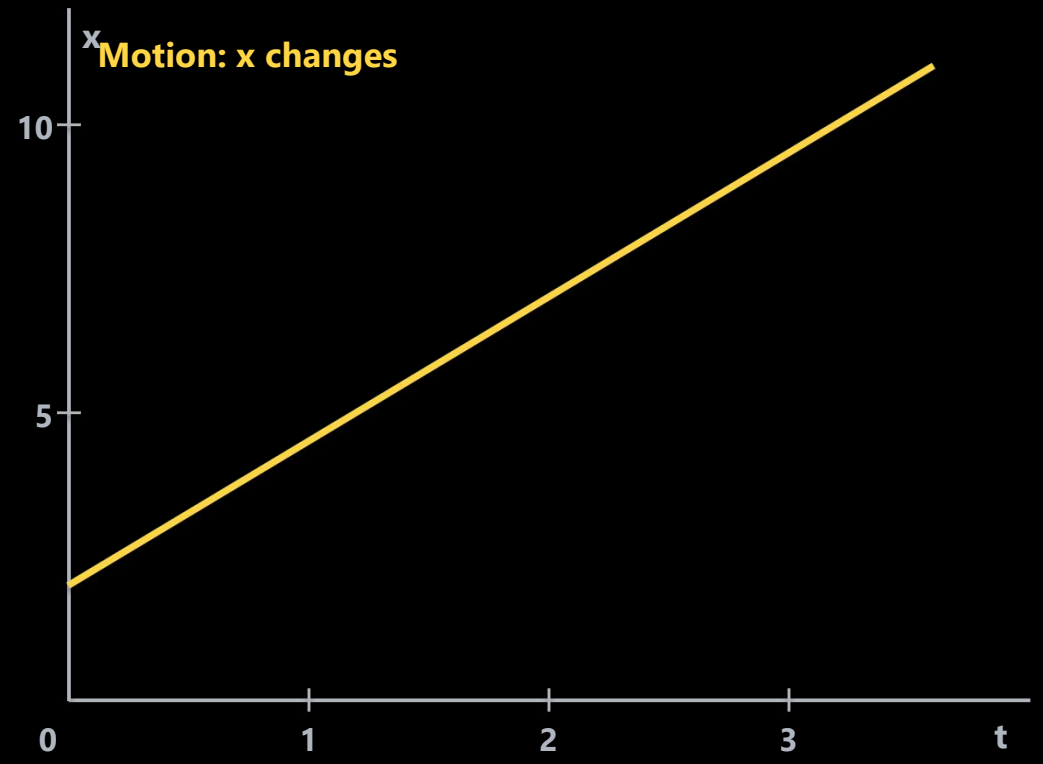
Position changes with time.
x moves to a new value.

Rest & motion are relative to the frame of reference

x–t picture: rest vs motion

On a position–time graph, rest is a horizontal line. Motion is a rising or falling line (or curve).





Rough work / extra work

Distance — the path you really walked

Distance is the length of the actual path covered by the object.
A vehicle odometer reading is distance.

- Scalar — only size, no direction
- Unit: metre (m)
- Cannot be negative
- Keeps increasing while the object moves
- Depends on the path between two points

If the path is unknown, you cannot calculate distance.

Odometer idea

Adds every metre of path

never goes backwards



long path = large distance

Displacement — the net shift

Displacement is how much the position shifted, and in which direction. It is the shortest directed jump from start to finish.

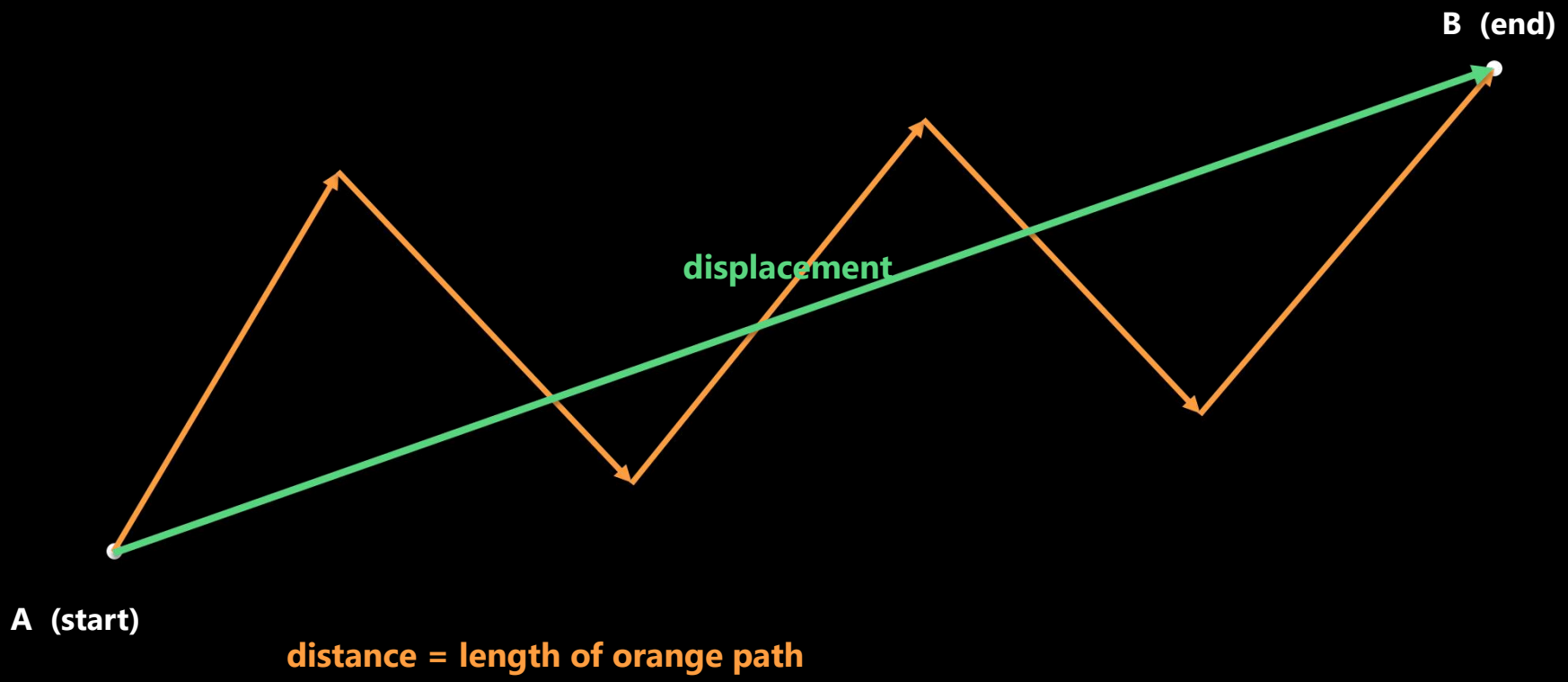
$$\Delta r = r_f - r_i$$

- Vector — size and direction; unit metre
- Does not depend on the path taken
- Can be positive or negative on a line (sign = direction)
- Can decrease even while the object is still moving

If initial and final positions are the same → displacement = 0

Path (distance) vs straight jump (displacement)

Same start A and end B. The winding route is distance. The straight arrow is displacement.



Three inequalities you must own

In one dimension, if the object never turns back, the size of displacement equals the distance. In every case, distance cannot be smaller than |displacement|.

No change of direction (1-D): |displacement| = distance

Always (any path): distance ≥ |displacement|

distance / |displacement| ≥ 1

|displacement| > distance is never possible.

MCQ — motion in a straight line

Which option is correct for motion in a straight line (1-D)?

1. distance = |displacement|
2. distance > |displacement|
3. distance ≥ |displacement|
4. distance < |displacement|

Solution: MCQ — motion in a straight line

Idea: without a turn, equal; with a turn, distance is larger. Both sit under ≥.

Step 1: Correct: distance ≥ |displacement|

Answer: option 3

Rough work / extra work

MCQ — numerical ratio

The numerical ratio of distance to |displacement| is always

1. less than one
2. equal to one
3. equal to or less than one
4. equal to or greater than one

Solution: MCQ — numerical ratio

Step 1: distance / |displacement| ≥ 1

Answer: equal to or greater than one

MCQ — 1-D without change of direction

For 1-D motion without change in direction, which is correct?

1. distance = |displacement|
2. distance > |displacement|
3. distance ≥ |displacement|
4. distance < |displacement|

Solution: MCQ — 1-D without change of direction

Idea: the object never turns, so path length equals net shift size.

Answer: distance = |displacement|

Rough work / extra work

MCQ — 1-D with change of direction

For 1-D motion with a change of direction, which is correct?



1. distance = |displacement|
2. distance > |displacement|
3. distance ≥ |displacement|
4. distance < |displacement|

Solution: MCQ — 1-D with change of direction

Answer: distance > |displacement|

Because it turned, the path is longer than the straight jump.

Number-line method

Mark start and end on a line. Distance = sum of every segment you walk. Displacement = final coordinate − initial coordinate (as a vector along î).

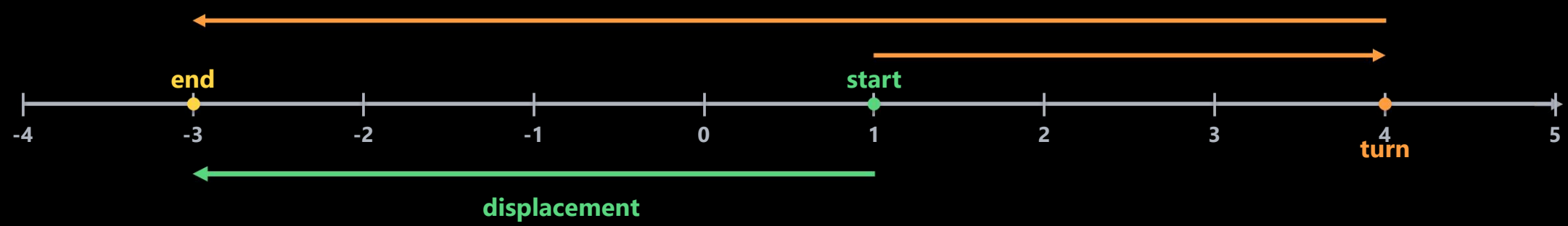
displacement = $r_f - r_i$ → on a line: $(x_f - x_i)\,\hat{i}$



① A at +1 → B at +5: distance = 4 m , displacement = +4 î

With a turn on the line

Start at +1 m, go to +4 m, then reverse and stop at −3 m.



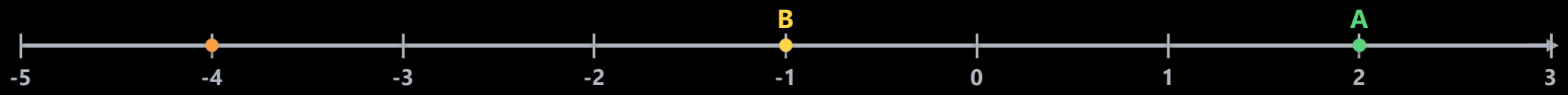
distance = 3 + 7 = 10 m

displacement = (−3)î − (1)î = −4 î

Two more line drills

③ Start +2 → go to −4 → end −1

distance = 6 + 3 = 9 m , displacement = (−1 − 2)î = −3 î



④ Start −3 → go to +4 → end −4

distance = 7 + 8 = 15 m , displacement = (−4 − (−3))î = −1 î

Rough work / extra work

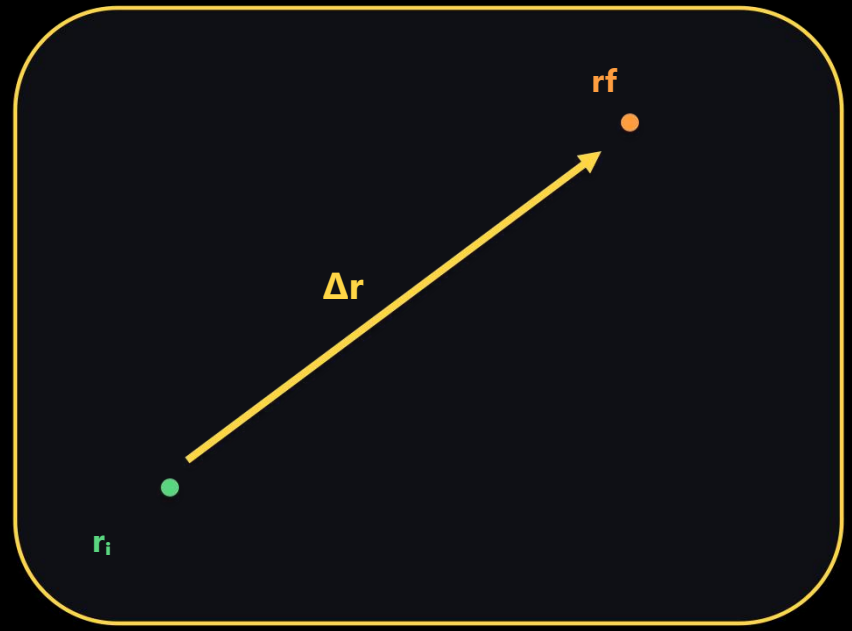
3-D shift of position

Initial (2, 4, 7), final (5, 8, 12). Find displacement.

$\Delta \mathbf{r} = (5-2)\hat{\imath} + (8-4)\hat{\jmath} + (12-7)\hat{k} = 3\hat{\imath} + 4\hat{\jmath} + 5\hat{k}$

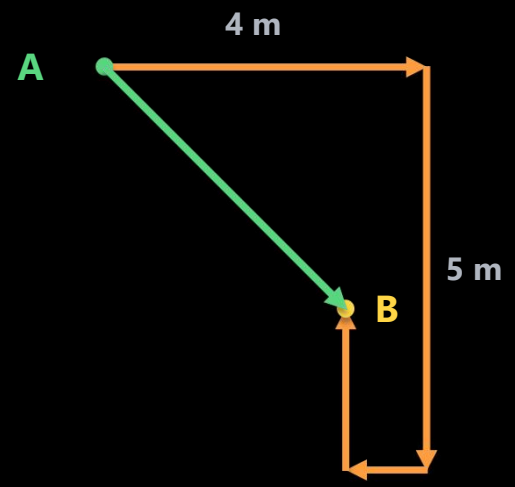
$|\Delta \mathbf{r}| = \sqrt{(9 + 16 + 25)} = \sqrt{50} = 5\sqrt{2}$ m

Displacement $3\hat{\imath} + 4\hat{\jmath} + 5\hat{k}$; magnitude $5\sqrt{2}$ m



Broken path A → B on a plane

Walk: 4 m right, 5 m down, 1 m left, 2 m up. Find distance and displacement.



distance = 4+5+1+2 = 12 m

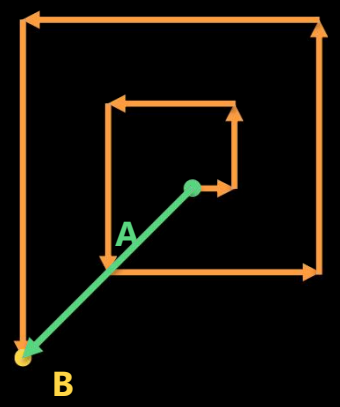
net: 3 m right, 3 m down

$\Delta r = 3\hat{i} - 3\hat{j}$

$|\Delta r| = 3\sqrt{2}$ m

Spiral path (segments 1 to 8)

Path lengths 1, 2, 3, 4, 5, 6, 7, 8 m in turn (right, up, left, down, …).



distance = 1+2+…+8 = 36 m

net forms 4–4 right triangle

|Δr| = √(16+16) = 4√2 m

Zero distance / zero displacement

If ...	Then ...
distance = 0	displacement must be 0 (did not move)
distance ≠ 0	displacement may be 0 or not
displacement = 0	distance may be 0 or not
displacement ≠ 0	distance must be > 0



A = B

disp = 0, distance > 0

True / False — lock these

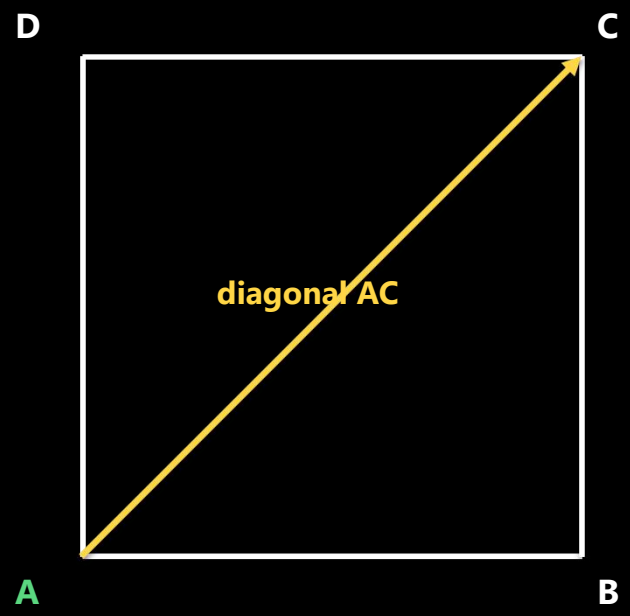
1. Distance must equal |displacement| in every 1-D motion. → False
2. If displacement is zero, distance must be zero. → False
3. If distance is zero, displacement must be zero. → True
4. |displacement| is always equal to distance. → False

Only when direction never changes in 1-D can we say |displacement| equals distance.

Rough work / extra work

Square park of side ℓ

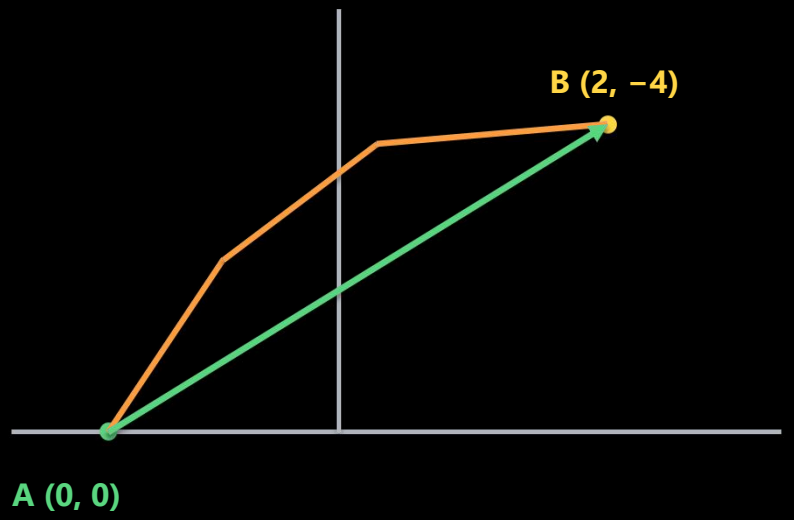
Start at corner A of square ABCD. Walk on the boundary. Side length = ℓ.



Path	distance	|disp
A → B	ℓ	ℓ
A → C (via B)	2ℓ	$\ell\sqrt{2}$
A → D (via B, C)	3ℓ	ℓ
A → A (full lap)	4ℓ	0

When distance cannot be found

Particle goes from (0, 0) to (2, −4) on a curved flight path. Without the exact path length, distance is unknown — displacement is still easy.

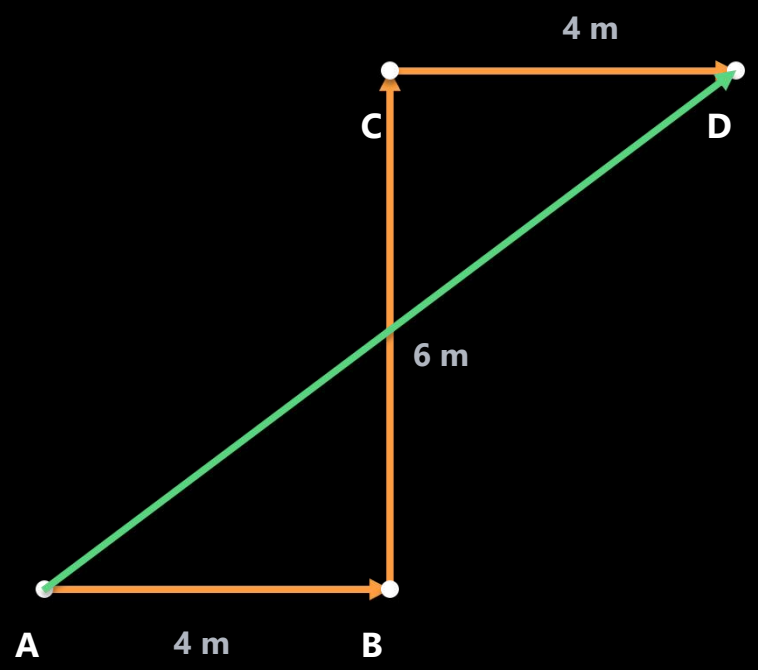


$\Delta \mathbf{r} = 2\hat{\imath} - 4\hat{\jmath}$, $|\Delta \mathbf{r}| = 2\sqrt{5}$

Straight drop (0, 0) → (0, −18): distance = 18 m and displacement = $-18\,\hat{\jmath}$.

L-shaped path on axes

A → B = 4 m right, B → C = 6 m up, C → D = 4 m right.



distance = 14 m

net: 8 m right, 6 m up

|Δr| = √(64+36) = 10 m

distance 14 m , |displacement| 10 m

MCQ — displacement P → Q

Particle moves from P(2, 3, 5) to Q(3, 4, 5). Displacement is

1. $\hat{i} + \hat{j} + 10\hat{k}$
2. $\hat{i} + \hat{j} + 5\hat{k}$
3. $\hat{i} + \hat{j}$
4. $2\hat{i} + 4\hat{j} + 6\hat{k}$

Solution: MCQ — displacement P → Q

Step 1: $\Delta\vec{r} = (3-2)\hat{i} + (4-3)\hat{j} + (5-5)\hat{k} = \hat{i} + \hat{j}$

Answer: $\hat{i} + \hat{j}$

Rough work / extra work

MCQ — correct statement

Correct statement among the following is:

1. When displacement is zero, distance travelled is not zero.
2. When displacement is zero, distance travelled is also zero.
3. When distance is zero, displacement is not zero.
4. Distance travelled and displacement are always equal.

Option 1 can happen (round trip) but is not always true. Option 3 is impossible.

Solution: MCQ — correct statement

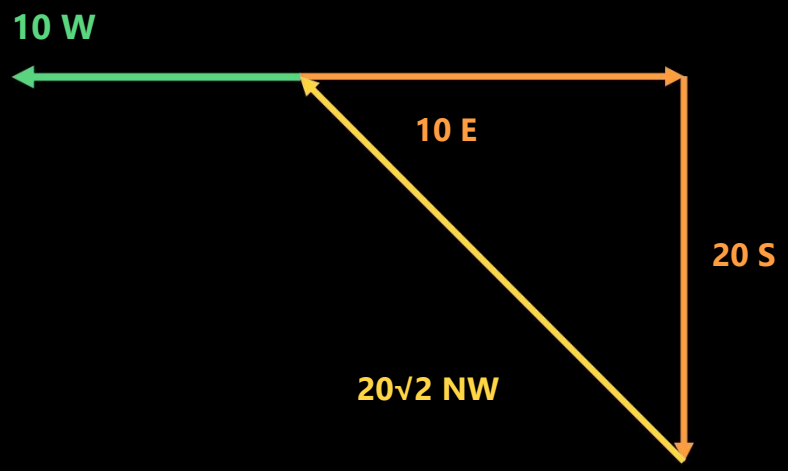
Idea: none of 2–4 are always true. The useful lock line is: displacement zero does not force distance zero.

Best teaching answer from the logic table: if displacement = 0, distance may or may not be zero. Statement 1 describes a possible case (return trip).

Focus: zero displacement ⇏ zero distance

East–South–North-West walk

Walk 10 m east, then 20 m south, then 20√2 m north-west. Find displacement.



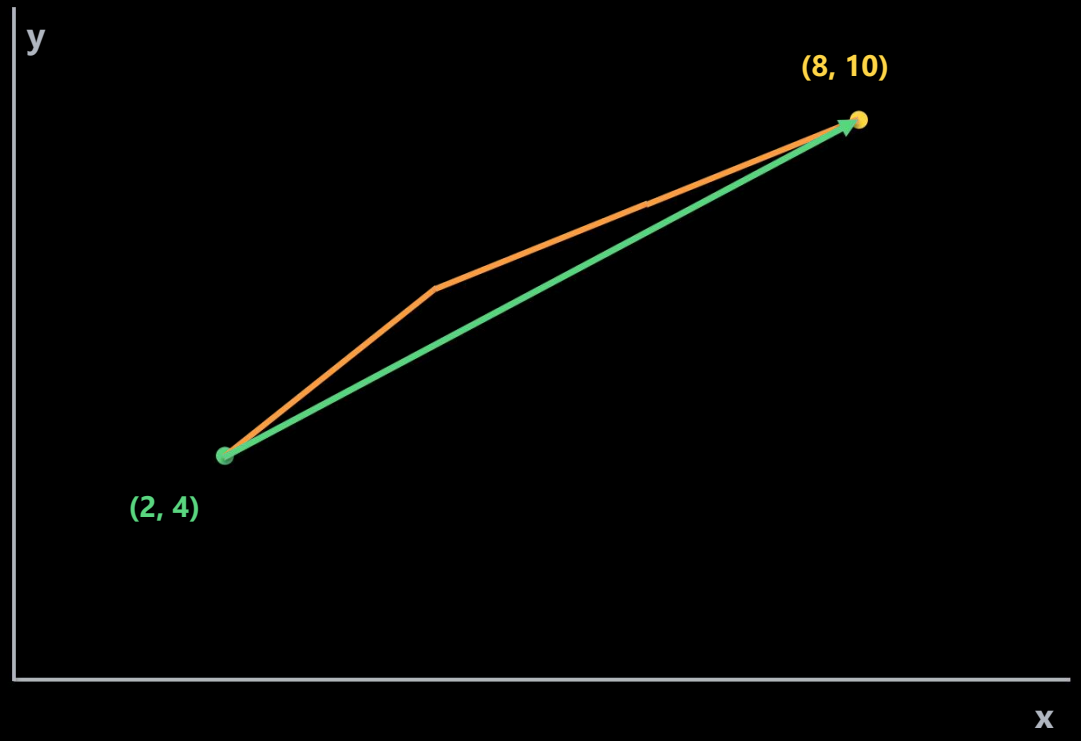
NW step = 20√2 × (−î + ĵ)/√2 = −20 î + 20 ĵ

Net = 10î − 20ĵ − 20î + 20ĵ = −10 î

Displacement = 10 m due west

Curved path on a slope (2, 4) → (8, 10)

Object moves from (2, 4) to (8, 10) along a curved path on a line. Path length may be unknown; displacement is not.



$\Delta r = (8\hat{i}+10\hat{j}) - (2\hat{i}+4\hat{j}) = 6\hat{i} + 6\hat{j}$

$|\Delta r| = 6\sqrt{2}$

Say these aloud

- Frame first — every observation needs a reference.
- Observer assumes he is at rest with respect to himself.
- Position changes with time → motion; else rest (in that frame).
- Distance = path length (scalar). Displacement = $r_f - r_i$ (vector).
- Always: distance ≥ |displacement|. Never the reverse.
- Same start and end → displacement = 0 (distance may still be large).
- No turn in 1-D → |displacement| = distance.

Next: deeper distance–displacement practice (PW 02).

Rough work / extra work

Thank you

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